

Expanded tile t (32 rows × 256 columns)

tile_colidx[ ]

3	17	42	88	131	202	245	⋯

distinct columns in t → stage feature rows once, reused by all 32 rows

per-row segments (local to the tile)

seg_ptr[ ] *row-start offsets*	0	2	5	5	9	⋯
seg_colidx[ ] *local column ids*	1	4	0	3	5	⋯
seg_values[ ] *rating values*	4.0	3.5	5.0	2.0	4.5	⋯

$d_t = \mathrm{nnz}(t)/(XB \cdot YB)$ — drives FP16/FP32 staging choice ($d_t \geq \tau_1 = 10^{-3}$)

compacted active-tile list (≈ 30% empty tiles removed)